Journal

We spoke through a design software when planning our project.

The First Few Weeks:

Planning on idea: Came to conclusion of making a gas detector for homes.

Planning what parts are needed

Handling the software of the model

(05/25)

Lubin - Hierdie is die website se naam waar ons die goed vir die gas detector kan koop: Micro Robotics. (Dit is in ZA en redelik cheap maar goeie kwaliteit)

General lys van goed wat ons gaan kort om dit te maak, ek sal later nog meer spesifiek se watse parte:

1. Gas Sensor (Carbon Monoxide & Flammable Gas Sensor MQ-9) - Detects harmful gasses:
   - MQ-9: R126
2. Microcontroller (ESP32) – Supports MicroPython for Python-based programming:
   - ESP32 Dev Board CH340 - USB-C: R98
3. LCD Display :
   - Wave 2.8" Capacitive Touch LCD Display
4. Buzzer or LED Indicator – Alerts when dangerous gas levels are detected:
   - AD16 Buzzer with Light - Red, 12V: R38
5. Water-Resistant Enclosure – Protects components from moisture:
   - GAINTA IP68 ABS Enclosure 121x171x55, with Wings: R215
6. Power Supply (12V DC Adapter) – Reliable voltage for gas detection systems:
   - Hi-Link AC/DC 12V 0.25A 3W Power Module: R78
7. Rechargeable Battery Pack (12V Li-ion or LiPo) – Provides backup power during outages:
   - Samsung 18650 3500mAh 3.7V Li-ion Battery: R240
8. **charger?:**
9. Battery Management Module LiPo Charger 3.7V with DC-DC Boost 4.5-24V - Type C
10. Voltage Regulator (Buck Converter) – Ensures stable voltage for components:
    - Voltage Regulator 7805 5V (2 Pack): R19
11. DC Barrel Jack (2.1mm / 5.5mm) – Standard connector for 12V adapters, compatible with microcontrollers:
    - DC Barrel Jack (2.1mm / 5.5mm): R21
12. JST Connectors – Useful for low-power connections between sensors and microcontrollers: JST-XH 2.54mm Connector Kit (560 Pieces): R296
13. Battery Connector Kit (50A or 120A) – Helps quick disconnection/reconnection of the backup battery:
    - Battery Connector Kit - 120A Anderson Compatible: R210
14. DC Jack to Alligator Clips – Converts a wall adapter into a mini power supply for easy testing:
    - DC Female Jack to Alligator Clips - 50cm: R25
15. Standard Electrical Wires (18-22 AWG for low power, 14-16 AWG for higher current) – Ensures proper power distribution:
    - Dual Red & Black Wire 22AWG - 5 meter: R64

TOTAL COST: R1630 (Ekt vergeet van taxes vir die eerste paar detectors so die prys is heelmoontlik nog hoer, dalk hier by 1800 of 1900)

Ek het gedink die gas detector kan konstant by n muurprop in geprop wees, maar dat hy ook n backup battery sal he vir as loadshedding of iets gebeur dat hy darem nog krag het. Dit gaan obv die production price op vat, maar ek dink dis n worthy safety precaution wat ons def kan extra punte score.

Ons kan eers probeer n custom case koop om al die komponente in te sit, maar later aan as ons sien dit werk nie of ons sukkel kan ons my oom vra om n case te 3d print. Enigste ding is dit moet dalk kan ligte hitte weerstaan omdat dit naby n stoof gaan wees en dalk ook waterproof moet wees omdat dit naby n stort kop gaan wees.

(05/26)

Sebastian - Dit klink goed👍Ek kyk nou na die coding en setup vir MicroPython. Ek gaan dit probeer begin (die coding) teen volegende naweek. Ek kort darem nie die fisiese dinge nog nie, maar ek gaan net die regte modelle moet weet.👌

Ek kort net die presiese model van die microcontroller asb🙏

Ek wil net ook vra, dink jy nie die gaan 'n biejtie te duur wees in terme van per eenhede prys vir die model? Ek neem aan van die dinge is seker in excess(die kits and whatnot)?

(06/01)

Lubin - Dis noggals duur maar ons sal maar kyk hoe werk hy en of ons kan cut back met enige iets. Die presiese microcontroller is die ESP 32👍

Sebastian - Dis reg.👌

Ek kyk deur die minipython program dit lyk nie te bad. Ek sal net moet die microcontroller se ins and outs moet leer👌

(06/22)

Lubin - Did a complete rework on our introduction, making it more appealing and making sure it is according to the guidelines. Also started and got far with the research plan.

(06/24)

Sebastian - Did some formatting of the project with headings and subheadings so work is more clear. Further research on the coding for the product has been done. Will test code with uPyCraftIDE and ESP32 microcontroller in coming days. It is best to use the ESP 32 Microcontroller for it supports microPython. The Arduino board is more accustomed for C and C++. It is possible to communicate with the board with the Firmata Library but it will be more complex. Will try both to test which is best. Further research and testing may be done on this in the future. A diagram is being made for the product. Determining which components are best.

(07/02)

Lubin - Perfected the introduction and started on the method section, stating the materials, procedure, ethics and safety precautions that will be needed for the testing of the gas detector

(07/03)

Sebastian - Updating the research plan's format and organization of project. Working on the diagram. Ordering necessary parts for project: ESP32 board, buzzer, breadboard for testing, m-m wires, transistor and resistor

(07/05)

Sebastian - Found out how to accurately measure gas with the gas detector. Going to use own gas stove with plastic bag (Propane) and other methods to measure CO. Going to order all parts on communica and DIY Electronica. Then assembling will begin. Refinements has been done on research plan.

Lubin - Did refinements on the research plan and report file, splitting them into two different documents and ensuring they both fulfill their respective needs according to the expo document. Also did refinements on the grammar, spelling and tenses of the research plan.

(07/07)

Sebastian - Made the first sketch diagram of how components will be powered in the circuit. More diagrams will be made as building goes on. Some websites don't have stock so complications came up.

(07/11)

Sebastian - All parts have been ordered and shipped. The Esp32 has already arrived. Need a micro usb charger to code with still. All parts expected to come monday.

(07/13)

Sebastian - Building has begun. Complications have come up. Like needing to know how to make a voltage divider. And I needed a multimeter.

(07/16)

Sebastian - Change of how the components will work and what was needed in the research plan. Roughly trying to update previous answers and questions in research plan. Such as the goal has changed to make a gas detector that can be used whilst the power is off during loadshedding. Placing a bigger importance on the battery.

(07/17)

Sebastian - Updating power diagram to phase 2--via micro. The esp23 can output both 3.3v and 5v if it is powered via its micro usb c port. So the buck converter only powers the esp32 itself now. and the esp32 is able to power all other components. Also added a wiring diagram of the current project state. It consists mainly of the esp32 and how it is wired with the gas sensors and buzzer.

(07/18)

Sebastian - going to test today. recorded results, trends are clear. The mq 5 seems to be more sensitive in general to all the gasses that were released. But it did still detect the harmful gasses.

(07/21)

Lubin - Made some major changes to the research plan to ensure the information is still valid and accurate to changes made to the gas detector.

(07/23)

Sebastian - Updated materials list. will make use of a charger for the 12v apg battery. this allows the apg battery to be charged with house power. while the battery is being charged it powers the buck converter with 12v, the buck converter then converts it to 5V. The buck converter can then supply the esp32 microcontroller through adapting the power with a micro usb c cable so we can charge it via that specific port for the esp32. this frees up the 5V VIN pin to be an output, where the voltage isnt regulated down to 3.3v. This allows the esp32 to power the gas sensors(mq5 and mq7). the esp32 can supply 5v and 3.3v. Thus it powers the vcc of both gas sensors via the v pin, for they need 5v. the 3.3v pin of the esp32 is used to power the buzzer because it needs both an gpio pin and vcc pin to be powered. The buzzer is an active low buzzer, meaning it powers on when the gpio pin need 0v for the buzzer to turn on. if the buzzer vcc pin is getting 5v from the esp32, the gpio would need 5v t power it off, which the esp32 can not control or output via a gpio pin, thus 3.3v is best to work with. when needed the esp32 sends a high or low voltage via a gpio pin to the buzzer's gpio to make it go on or off. the gas sensors work with 5v, they have an analog ouput and digital output.which can range from 0V to 5v. the esp32 can only recieve 3.3v via its gpio pins, so a voltage divider will be needed. Also updated the procedure according to what was best.

(07/24)

Sebastian - Added information about data analysis- how things will be measured.

Lubin - Added the limitations and errors, and recommendations for future reference.

(07/27)

Sebastian - Completed poster notes and abstract.

Lubin - Completed poster notes.

(07/28)

Lubin - Did final grammar and spelling checks.

Sebastian - Did final fact checking and data comparisons.